

Engineering Project Report

Simple Joule Thief Circuit: Low-Power Inductive Voltage Booster

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Abstract

This report documents the design, electromagnetic analysis, and laboratory evaluation of a Joule Thief self-oscillating voltage booster circuit. The Joule Thief is a minimalist flyback boost converter designed to extract residual energy from depleted chemical batteries ($0.8\text{ V} - 1.1\text{ V}$). By utilizing a dual-winding (bifilar) toroidal inductor, an NPN switching transistor (2N2222), and a single base resistor, the circuit generates high-frequency voltage spikes ($3\text{ V} - 15\text{ V}$ at $50\text{ kHz} - 500\text{ kHz}$) capable of driving standard LEDs ($V_F \approx 3.0\text{ V}$). This document presents the five-phase self-oscillation cycle, inductor saturation mechanics, winding rules, component selection criteria, and practical performance limitations.

Contents

1 Introduction & Operating Concept

Conventional light-emitting diodes (LEDs) require a minimum forward threshold voltage ($V_F \approx 2.0\text{ V} - 3.4\text{ V}$) to initiate photon emission. Standard 1.5 V AA/AAA alkaline cells are typically discarded once their output voltage drops below 1.1 V, even though significant chemical energy remains unextracted.

The **Joule Thief** overcomes this voltage threshold by using inductive energy storage and rapid transistor switching, boosting input voltages as low as 0.7 V – 0.8 V into high-voltage pulses.

2 Bill of Materials

Table 1: Bill of Materials for Joule Thief Circuit

S.No.	Component	Qty	Part Number	Circuit Function
1	NPN Transistor	1	2N2222 / 2N3904	High-speed switching device
2	Toroidal Core	1	Ferrite Ring Core	Magnetic flux storage medium
3	Magnet Wire	2 Strands	28 AWG Enamelled	Primary & Feedback bifilar windings (10 – 15 turns)
4	Base Resistor	1	1 k Ω , 1/4W	Limits base drive current and sets frequency
5	Output Load	1	5mm White LED	Light output indicator ($V_F \approx 3.2\text{ V}$)
6	Power Source	1	1.5 V AA Battery	Depleted cell (0.8 V – 1.2 V)

3 Toroid Winding Architecture

Proper phase alignment of the two windings is critical for positive feedback:

1. Wind two insulated copper wires (Red = Primary, Yellow = Feedback) simultaneously around the ferrite toroid for 10 – 15 turns (**Bifilar Winding**).
2. Connect the finish of the Primary winding (R_2) to the start of the Feedback winding (Y_1). This junction forms the **Center Tap** connected to $+V_{\text{battery}}$.
3. Connect the start of the Primary winding (R_1) to the transistor **Collector**.
4. Connect the finish of the Feedback winding (Y_2) through the 1 k Ω resistor to the transistor **Base**.

4 Five-Phase Oscillation Dynamics

4.1 Phase 1: Initial Base Conduction

Current flows from the battery through the Feedback winding and 1 k Ω resistor into the Base, turning the transistor slightly ON. Current begins flowing through the Primary winding into the Collector.

4.2 Phase 2: Regenerative Positive Feedback

Primary current induces a positive voltage in the Feedback winding via mutual inductance. This increases Base current, rapidly saturating the transistor into full conduction.

4.3 Phase 3: Magnetic Core Saturation

Primary current increases until the ferrite core reaches magnetic saturation ($\frac{d\Phi}{dt} \rightarrow 0$). Induced voltage in the feedback winding collapses to zero, cutting off base current and turning the transistor OFF abruptly.

4.4 Phase 4: Inductive Flyback Voltage Spike

With the transistor OFF, the collapsing magnetic field generates a high-voltage back-EMF pulse according to Faraday's Law:

$$V_L = -L \frac{di}{dt} \quad (1)$$

This voltage spike (3 V – 15 V) adds to the battery voltage, exceeding the LED forward voltage threshold (V_F) and lighting the LED.

4.5 Phase 5: Cycle Reset

Once magnetic energy is depleted into the LED, the coil voltage drops to zero. Small battery current flows back into the base, initiating Phase 1 again at 50 kHz – 500 kHz.

5 Performance Summary & Applications

- **Minimum Input Voltage:** Runs reliably down to ~ 0.75 V.
- **Oscillation Frequency:** 50 kHz – 500 kHz depending on core permeability and turns ratio.
- **Applications:** Emergency flashlights from dead batteries, micro-energy harvesting, and inductive switching educational kits.

6 Conclusion

The Joule Thief provides a simple, self-oscillating demonstration of energy extraction from low-voltage sources, transforming weak battery energy into bright LED illumination via flyback induction.